

BASE SAN MARTIN, Antarctica

WMO: 890660

Lat: 68.130S

Lon: 67.104W

Elev: 23

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-21.7	-15.7	-31.4	0.9	-19.8	-24.4	1.4	-15.3	90.5	3.1	67.9	6.5	2.9	140	1.026

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	43.6	36.9	42.1	35.8	40.9	34.9	37.8	42.1	36.7	40.7	35.8	39.5	11.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.4	29.1	37.4	33.3	27.9	36.3	32.5	27.0	35.7	14.4	42.3	13.9	41.0	13.5	39.8	44.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 63.6	49.5	39.6	-24.5	47.7	7.8	2.0	-30.1	49.1	-34.7	50.3	-39.1	51.4	-44.7	52.8		
(5)			-24.8	41.1	7.2	1.6	-30.0	42.2	-34.3	43.1	-38.3	44.0	-43.6	45.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	35.3	33.7	30.7	25.9	24.7	16.6	10.5	11.4	15.8	23.1	28.3	33.4		
	DBStd	12.13	3.01	3.50	4.88	6.21	7.57	10.46	13.47	13.53	13.04	9.14	6.22	3.51		
	HDD50	9465	457	455	597	723	785	1003	1223	1196	1025	834	652	515		
	HDD65	14940	922	875	1062	1173	1250	1453	1688	1661	1475	1299	1102	980		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.6	9.0	9.8	10.7	12.2	10.7	11.0	9.8	10.5	10.9	11.3	11.3	10.0		
	PrecAvg	27.0	2.1	2.2	2.5	2.2	2.2	1.8	1.9	2.2	2.6	2.9	2.6	1.8		
(15) Precipitation	PrecMax	39.8	3.9	3.7	4.3	4.3	4.4	3.3	3.4	3.6	3.6	5.3	4.2	4.3		
	PrecMin	18.2	1.1	1.2	1.4	0.6	1.3	0.8	0.7	0.6	1.6	1.5	0.9	0.4		
	PrecStd	5.6	0.9	0.7	0.6	0.8	0.7	0.7	0.6	0.7	0.6	0.9	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	44.5	44.2	42.6	43.1	40.3	40.6	39.0	42.2	43.1	43.3	44.6		
		MCWB	38.5	37.5	38.4	36.8	37.4	34.8	34.5	34.3	36.0	37.9	36.7	37.1		
	2%	DB	42.8	42.2	41.2	39.1	39.5	36.4	36.5	35.4	39.1	40.4	40.4	42.0		
		MCWB	36.0	35.5	35.9	33.8	34.7	32.6	32.3	32.5	34.1	35.4	34.5	35.3		
	5%	DB	41.3	40.5	39.0	36.0	37.1	32.3	32.2	32.5	36.4	38.2	38.3	40.4		
		MCWB	35.0	34.6	33.9	32.3	32.7	30.7	29.5	30.3	32.6	33.9	33.3	34.2		
	10%	DB	40.0	39.0	37.0	33.9	34.5	29.7	28.6	29.4	33.2	35.5	36.2	38.8		
		MCWB	34.3	33.7	32.9	31.3	31.4	28.2	26.9	27.7	30.7	32.2	32.3	33.2		
	0.4%	WB	39.8	38.5	39.1	37.2	37.6	35.7	35.0	35.1	36.6	38.7	37.5	38.0		
		MCDB	44.3	43.6	43.4	41.9	41.8	39.2	39.4	37.2	40.5	42.3	42.0	43.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	36.5	36.4	34.5	35.2	32.8	32.4	32.6	34.6	36.0	35.4	36.2		
		MCDB	41.1	40.3	40.2	38.0	38.7	35.4	36.0	34.9	38.4	39.7	39.1	40.7		
	5%	WB	36.0	35.5	34.7	32.9	33.2	30.8	29.5	30.4	32.8	34.2	34.0	35.1		
		MCDB	39.6	39.1	37.8	35.3	36.2	32.5	31.8	32.4	36.0	37.6	37.4	39.1		
	10%	WB	35.2	34.5	33.5	31.5	31.5	28.4	26.8	27.7	30.8	32.2	32.7	34.0		
		MCDB	38.6	37.7	36.1	33.5	34.1	29.5	28.5	29.3	33.2	35.2	35.8	37.5		
	5% DB	MDBR	6.3	6.2	7.0	7.0	7.7	9.6	12.4	13.6	13.2	11.2	9.1	6.9		
		MCDBR	8.5	8.5	9.6	9.4	10.1	11.2	14.0	14.9	13.0	12.3	10.1	8.9		
		MCWBR	5.5	5.4	6.6	6.6	7.1	9.2	12.2	13.0	9.9	9.3	6.9	5.7		
		MCWBR	7.4	7.5	8.6	8.7	9.4	10.8	14.3	14.2	12.9	12.1	10.0	8.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.3	5.3	6.4	6.5	7.4	9.3	12.8	12.7	10.5	9.8	7.7	5.9		
	Clear-Sky Solar Irradiance	taub	0.269	0.263	0.243	0.201	N/A	N/A	0.140	0.196	0.236	0.251	0.259	0.269		
		taud	2.328	2.361	2.366	2.219	N/A	N/A	1.908	2.098	2.224	2.226	2.255	2.273		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	312	295	267	200	N/A	0	43	191	261	295	313	317		
		Edh at Noon	34	28	21	12	N/A	0	2	13	23	32	36	37		
(44) All-Sky Solar Radiation	RadStd		2287	1505	726	203	24	0	7	121	514	1330	2160	2641		
			100	98	62	20	2	0	1	9	46	75	99	131		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page